Predicting NFL Game Outcomes Using ELO Ratings and Machine Learning Models

1st Sai Sudeshna Vallamkondu
Department of Information Science,
University of North Texas
Denton, USA
saisudeshnavallamkondu@my.unt.edu
11642147

2nd Arun Kumar Machupally
Department of Information Science,
University of North Texas
Denton, USA
arunkumarmachupally@my.unt.edu
11640024

3rd Saikumar Dumpoju
Department of Information Science,
University of North Texas
Denton, USA
saikumardumpoju@my.unt.edu
11654851

4th Ranganadha Pawan Parankusam
Department of Information Science,
University of North Texas
Denton, USA
PawanParankusam@my.unt.edu
11594074

***Abstract*—This work carries out a comprehensive approach to forecast the outcomes of NFL games, achieved through the use of both an ELO rating system and machine learning algorithms. The dataset we forecast on consists of 2,510 regular season games played between 2009 and 2018. Post-hoc features such as scores, turnovers, and other performance indicators were engineered to make our predictions more accurate. Two machine learning models—Random Forest and Recurrent Neural Network (RNN)—were used to predict game outcomes based on the pre-game averages of the previous ten games. In addition to using machine learning, we created an ELO rating system that uses a Neo4j graph database to track team performance dynamically over time. When we tested our final predictions against real outcomes, we achieved an accuracy of about 63% over the last 500 games.**

Keywords: Random Forest, RNN, ELO rating system, Neo4j ,Machine Learning, Feature Engineering

I. Introduction to the Problem Statement

It is hard to predict the outcomes of NFL games because of the sport's complexity and the many things that can affect the results. This project attempts to predict the outcomes of the games using both the ELO rating system that uses the graph database to track performance and also we used some machine learning methods. The ELO system is a way of rating the teams that allows for a lot of real-time adjusting. It changes the ratings based on the results of the games,This methodology was essential for accurate reflection of dynamic team performance.

A. *Play-by-Play Structure of NFL Games*

A game in the National Football League [4] is split into four 15-minute quarters, but if the game is tied at the end of regulation, it can go into overtime to determine a winner. Throughout the game, teams rotate between offense and defense, with the goal of advancing the ball into the other end zone to earn points. Each play in an NFL game is made up of numerous important elements that are captured in play-by-play statistics. These variables include:

- Down: Number of attempts to move ball 10 yards
- Yards Gained: The number of yards a team moved the ball during a play.
- Play Type: Whether the play was a pass, run, or special teams play (e.g., punt or field goal attempt).
- Turnovers: Whether the ball was lost to the opposing team during the play.
- Time of Possession: The amount of time the ball was in team control during that play.
- Penalties: Infractions committed by players that result in yardage penalties or loss of downs.

When trying to predict game outcomes, we look at several different dimensions of what is happening on the field. What is happening over the course of the entire game, not just in isolated plays. For instance:

- Turnover Margin: How many turnovers is a team giving up? How many is it forcing? This seems like a good baseline for something that strongly influences game outcomes.
- Time of Possession: The longer a team has the ball, the more chances it has to score.This can be good indicator for the team performance.
- Total Yards Gained: The sum of the offensive production of a team seems like a good baseline for measuring how well that offense is doing.

When we combine the play-by-play details with game-

level metrics, they become variables that we can use in ELO calculations and machine learning algorithms. They matter because they represent details of overall team performance—performance that's crucial in making predictions about subsequent events.

B. *Relevance to Our Prediction Task*

In this project, We are not focusing on individual game plays but on the overall outcome of each game. Understanding of individual play contribution to a team's overall performance is critical for feature engineering. For instance:

- Turnovers: Teams that commit fewer turnovers tend to be more successful as they score more.
- Time of Possession: Teams that control the ball more often than not tend to be successful, too, because they keep their offense on field.
- Yards Gained: These are the kinds of variables at the play level that we convert to metrics at the game level.

By converting these play-level variables into game-level metrics, We use those metrics in both ELO and machine learning models to judge the team performance's direct effect on our ability to correctly forecast an individual team's future performance.

II. Literature Review

In this project, we employ an ELO rating system that closely resembles the methodology for ranking chess players from paper [7]. The ELO rating system for this project starts with a fixed set of initial ratings for all teams and then iterates through out the season, modifying the ratings based on the expected outcome of each game versus the actual game outcome. we look at the current team ratings and use them as the basis for forecasting the outcomes of the games. Assessing team performance in football over time for the use of the ELO system, which has been applied to many sports as it is known for its effectiveness. Not only it be used for the basic performance assessment, but also be relied upon for flexible,adjustments to your team strength ratings as new data becomes available. The ELO system involves raising or lowering a team's rating based on whether they won or lost the game , and how good the opposition team. In our method, we keep track of ELO ratings for each team in a graph database called Neo4j [3]. Each team in the NFL is represented as a node in the database, and each game serves as a connection (edge) between two nodes (teams). This edge captures relevant metrics about the game, like what the final score was and when the game happened. After a game is played, the ELO ratings for both teams are recalculated based on how well each team performed relative to our expectations. We use a simple formula, which will explain in detail, to carry out the operation.

$$E_A = \frac{1}{1 + 10^{\frac{R_B - R_A}{400}}}$$

where:

- EA is the expected probability of team A winning
- RA and RB are the current ratings of teams A and B respectively

The rating update formula used is:

$$R'_A = R_A + K(S_A - E_A)$$

where:

- K is the k-factor (set to 20 in our implementation)
- SA is the actual score (1 for win, 0.5 for tie, 0 for loss)
- EA is the expected score

The technique based on Neo4j provides efficient storage and retrieval of team ratings, with potential for real-time updates after each game. This choice was inspired from similar implementations in the sports analytics literature that uses the graph databases to model the relationship between teams and their games [2].

In addition to using ELO to rate the teams, we also used machine learning models like Random Forest and Recurrent Neural Networks (RNNs) to predict the outcomes of the games. These models have been shown to perform quite well in sports prediction tasks, primarily because they can identify important patterns in the data, statistical methods might miss. Thus, the core of our implementation derives from the well-established methods linked to ELO and from those related machine learning models that are used for predicting outcomes.

III. Objectives of the Study

The main aim of this project is to predict NFL game outcomes using both ELO rating methods and current machine learning techniques. The specific objectives listed below shows how we are proceeding:

- Implementing an ELO Rating System: Create an ELO rating system, using a Neo4j graph database, to track team performance over time. The database will updates the team ratings after each game and will facilitate dynamic adjustments in team strength based on actual versus expected performance across multiple games.
- Predict NFL Game Outcomes Using Machine Learning: Machine learning models have been employed to predict the results of NFL games based on a rich historical dataset. The models, which utilize features derived from the last 10 games, make use of metrics such as average scores, average turnovers, and average time of possession.
- Evaluate Prediction Accuracy: Evaluate the accuracy of the predictions made by both the ELO rating system and machine learning models. The accuracy will be measured by comparing predicted outcomes with actual results from the last 500 games in the dataset.
- Compare Feature Sets: Set different features against each other to see how effectively they work in improving the performance of the model. For instance, set the top 10 features against the top 25 features and see which one elevates the model's performance more significantly. This is important because it likely will give direct insight into which features matter most.

IV. Data Collection

This research utilizes a dataset of regular-season NFL games from 2009 to 2018. The data is taken from a service that reports the outcome of every play in "real-time" during a game, as well as provides a rundown of any key events that occur in the game (e.g., a coach's challenge, a penalty being enforced, etc.). The dataset captures a total of 2,510 games—each play in every game as well as certain key events that occur during a game.

A. *Data Description*

Each record in the dataset is a play which includes the following key variables:

- Game ID: A unique identifier for each game.
- Home Team:The name of the team playing at home.
- Away Team :The name of the team playing away.
- Home Score :The total points scored by the home team.
- Away Score :The total points scored by the away team.
- Game Date :The date when the game was played.
- Turnovers: The number of times each team lost possession of the ball due to interceptions.
- Passing Completions: The number of successful pass completions made by each team.
- First Downs Due to Penalties: The number of first downs each team received due to penalties committed by the opposing team.
- Time of Possession: The amount of time each team controlled the ball during a drive. etc.
  There are 255 characteristics play-by-play in the raw data of which 49 important characteristics per team were derived from the play-by-play data for each game.

V. Data Preprocessing: Calculating Features for Last 10 Games Averages from Raw Data

To compute the last 10 games averages for features from the raw NFL play-by-play data, we undertake a multidisciplinary approach. This entails not just cleaning the data and transforming it into a suitable format but also deriving key metrics through feature engineering. We describe this elaborate process step by step, including moments when we had to discard some games from our already reduced sample due to a lack of necessary historical context and when we had to make crucial decisions about what time window would be most useful for our machine learning and deep learning models.

A. *Data Acquisition and Preparation*

The raw data used in this study consists of detailed play-by-play information for each NFL game played between 2009 and 2018. Each row in the dataset represents a single play within a game, with variables such as:

- Game ID: A unique identifier for each game.
- Home Team:The name of the team playing at home.
- Away Team :The name of the team playing away.
- Play Type: Whether the play was a pass, run, or special teams play.
- Yards Gained: The number of yards gained on each play.
- Down: The down number during the play (1st, 2nd, 3rd, or 4th down).
- Turnovers: Whether the ball was lost to the opposing team.
- Time of Possession: How long the ball was in the team control.
  These variables are aggregated at the game level to create summary statistics to get the team performance for individual games.

B. *Transformation to Game-Level Metrics*

The play-by-play data was grouped into various categories such as offensive metrics, defensive metrics, and special teams metrics. These categories help us to understand how teams was performing. For example:

- Offensive Metrics: Include total yards gained, passing completions and time of possession.
- Defensive Metrics: Include turnovers forced and sacks recorded.
- Special Teams Metrics: Include how many field goals made and punt returns.

By aggregating all the above play-by-play metrics into game-level statistics as done by [1], we can create summary features that reflect overall team performance for each game. For example:

- Total Yards Gained: By doing sum of all yards gained by a team during the game gives a value of offensive performance.
- Turnover Margin: The difference between turnovers committed and turnovers forced by a team as it can be used as a strong predictor of game outcomes.
- Time of Possession: Teams that hold onto the ball longer tend to take control of games, keeping their defense while giving their offense more opportunities to score. These aggregated metrics are then used to calculate post-priori characteristics of the game.

C. *Calculating Post-Priori Characteristics*

Once we have combined the raw data of the game-level metrics, we calculate characteristics that summarize team performance over the last 10 games. These characteristics helps us to understand how well the team is performing:

- Average Points Scored (Last 10 Games): The average points scored by a team in their last 10 games.
- Average Points Allowed (Last 10 Games): The average points allowed by a team in their last 10 games.
- Turnover Margin (Last 10 Games): The difference between turnovers committed and turnovers forced by a team over their last 10 games.
- Time of Possession (Last 10 Games): The average time of possession for a team over their last 10 games.
  The previously discussed post-priori characteristics of each team are nothing more than the averages of the game-level metrics determined earlier. When all the characteristics are taken together, these averages offer how

each team is performing across the 10 games that were analyzed. By averaging the performance metrics over the last 10 games, we capture trends in the recent games.

D. *Dropping Games with Insufficient Historical Data*

Calculating the last 10 games averages for all teams is not easy, particularly in the early part of a given season. For instance, in Week 2 of the 2009 NFL season, most of the teams did not have 10 complete games in their historical data. In this case, 2009 historical data would not be used for either of the two affected teams because as we have incomplete data thus, invalid last 10 games averages for them. This step ensures that our machine learning models are trained on complete data without any missing values for key features.

E. *Feature Selection and Engineering*

After calculating the post-priori characteristics, we select the top features based on their correlation with the game outcome. These main features include:

- Points Scored (Last 10 Games): Teams that consistently score more points tend to have more wins.
- Points Allowed (Last 10 Games): Teams that allow less points tend to have stronger defense and they win more number of games.
- Turnover Margin (Last 10 Games): Teams with positive turnover margins have more scoring opportunities and are more chance to win.
- Time of Possession (Last 10 Games): Teams that control possession for longer periods often dominate games by keeping their defense rested and their offense on the field. These features are used as inputs for both ELO rating calculation and machine learning models.

Why Choose Last 10 Games Averages?

We have chosen averages from the last 10 games as features to predict future outcomes. As we know the last matches have more impact how we perform now. For example, [5] demonstrated that recent performance is a strong predictor of future outcomes. This appears to be an truth most of the time in sports analytics, and it applies well to football as a high-variability sport. When using last 10-game samples, one is largely confined to the current season. But the 10-game average speaks to both the current form and the overall season-average context.

Our approach involves averaging the last 10 games played by each team, which provides a clear enough picture of a team's current performance and also makes for a reliable historical context when it comes to predicting future game outcomes.

VI. Exploratory Data Analysis Insights

Exploratory Data Analysis (EDA) will be central to assessing relations between features within the data and how such relations impact game results. In this project,several visualization images were developed, which would show the distribution of key features and the degree of their association with the outcome of the games, as well as the possibility of such features as predictors.

A. *Scatter Plots: Home scores and Away Scores vs. Game Result*

- Home Score vs. Game Result: The scatter plot in the below picture shows that the relationship between the home team scores over the last 10 games ($last_10_games_home_score$) vs game result ($result$), where 1 represents a home team win, 0 represents a match tie, and -1 represents an away team win.
  Key Observation: At home matches, more points are usually scored by the teams which are in the winning side. Whenever a certain team attains a lot of points during home ground, that team is likely to win the game. In minority cases, when one team has a low score but still acceptable score it loses the match while the tie is within a more closer scoreline.

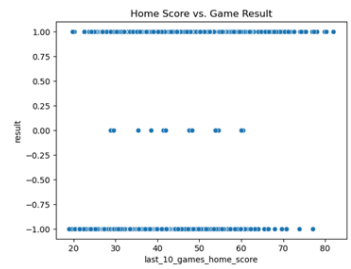

Fig. 1. Home Score vs Game Result.

- Away Score vs. Game Result: Similar to the home scores, the scatter plot for away scores in the figure ($last_10_games_away_score$) shows that higher away scores are associated with away wins (result = -1), while lower away scores are more likely to result in home wins (result = 1). Key Observation: Scores from away teams tend to be more varied than those from home teams, but this is not necessarily a bad thing for teams traveling to play in someone else's stadium. An away score is usually a good sign that a team is doing well, and doing even better than its usual home scoring rate.

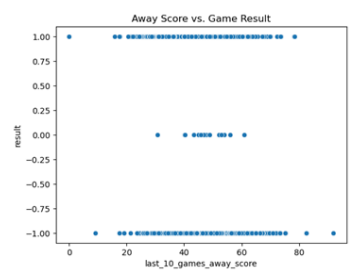

Fig. 2. Away Score vs Game Result.

B. *Box Plots: Score Distribution by Game Outcome*

The distribution of home team scores is compared across various game outcomes (home win, tie, away win) in the box plot. The main point is that winning home teams score more, on average, than teams that lose or tie. The IQR (interquartile

range), which shows the spread of middle scores, is also wider for home wins. The types of scores both above and below the median are more varied, which means winning home teams can score a much larger range of points in any given game. There are high-scoring games where the home team wins.

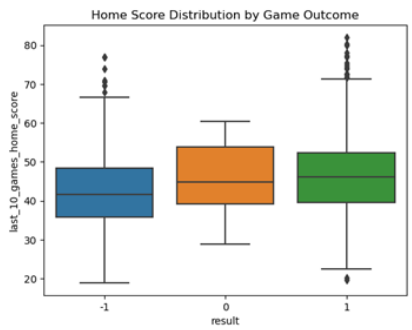

Fig. 3. Home Score Distribution by Game Outcome.

Distribution of Away Team Scores by Game Outcome: The box plot for away team scores displays a similar pattern. Winning away teams generally score more points, and losing or tying teams score fewer. A key observation is that away teams that win have a significantly higher median score than those that lose or tie. What's more, the interquartile range for scoring in away wins is less than that for home wins, suggesting that this year's away winners score more consistently than home winners. Outliers are problematic for our understanding of away scores. A couple of our worst scoring games have happened to away loser teams.

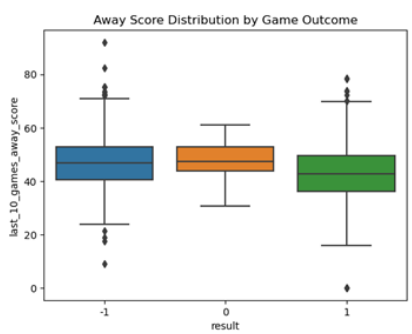

Fig. 4. Away Score Distribution by Game Outcome.

C. *Distribution of Home Score and Away Score*

- The KDE plot compares the distribution of scores for home teams ($last_10_games_home_score$) and away teams ($last_10_games_away_score$).
- Key Observations:
  - The distributions for both home and away scores are very similar, there is only little difference that indicates the home team score are slightly more than away teams.
  - From the figure we can describe that distribution of home and away are centered around a mean score of approximately 40 points, with a slight skew toward higher scores for home teams.
  - The ending of the distributions indicate that extreme high or low scores are low but from the figure it shows it occur for both home and away teams.

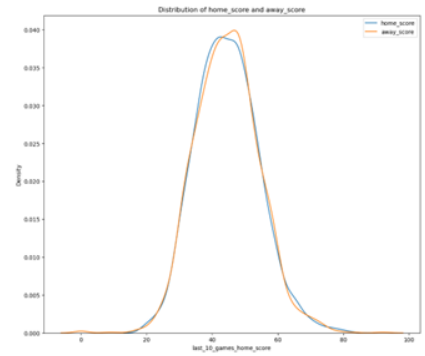

Fig. 5. Distribution of Home score and Away score.

D. *Results Distribution*

- The pie chart shows the distribution of game outcomes: percentage of home win, percentage of away win, and tie.
- Key Observations:
  - Home Wins Dominate: As we see that Home wins account for 57.1% of all games, which supports our hypothesis that there is little advantage for home teams.
  - Away Wins: Away wins make up 42.3% of all games, as there is advantage for home teams but there is not any significant loss for away teams as it also have good set of proportion.
  - Ties Are Rare: Ties account for only 0.6% of all games.

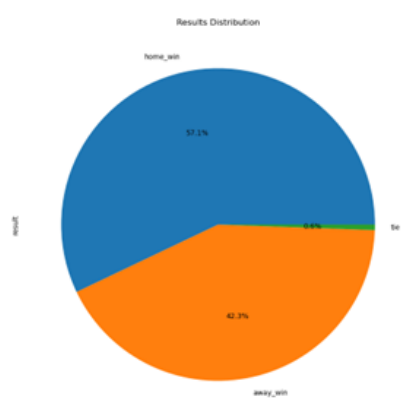

Fig. 6. Results Distribution.

E. *Top 10 Correlating Features to Result*

- The bar chart shows the top 10 features that have the highest correlation with the game result (win, loss, or tie).

- Key Observations:
  - The last_10_games_home_score has a strong correlation with the game result, which is expected since teams that score more points are more likely to win.
  - Quarter-Specific Scores: Features like last_10_games_home_score_q4 (home team's score in the 4th quarter) and last_10_games_home_score_last_2_minutes_q4 (home team's score in the last two minutes of the 4th quarter) are also highly correlated with game outcomes. This suggests that performance in critical moments of the game, especially late-game scoring, plays a significant role in determining the outcome.
  - Wins and Momentum: The feature last_10_games_home_number_of_wins also shows a strong correlation, indicating that teams with recent wins tend to maintain momentum and perform better in subsequent games.
  - Defensive Metrics: Features like last_10_games_home_score_q3_allow (points allowed by the home team in the 3rd quarter) and last_10_games_home_third_down_conv_perc (third down conversion percentage) show moderate correlations. This suggests that defensive strength and efficiency on critical downs are important factors for winning games.

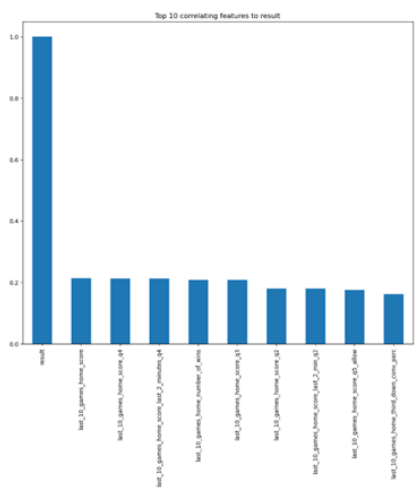

Fig. 7. Top 10 correlating features of result.

F. *Additional Insights from EDA*

Based on these visualizations, several insights can be drawn:

- Score Disparity and Outcome: The scatter and box plots confirm that not just scoring, but the way one scores (as a home or away team), matters when it comes to winning outcomes. If a team only barely manages to win, or if it scores below a certain threshold, then the appearance is that offensive play wins games—that, or a last-second defensive stop.
- Variability in Away Team Performance: Away teams exhibit a lot more variability in their performances and the scores they achieve than home teams do. One inference here is that home teams are more "stable" and consistent, though that's not something backed up with a statistical test. The scatter plots tell us that some away teams are capable of scoring at a clip that can win games; they also tell us that some away teams struggle much more than we ever see any home team struggle.
- Critical Performance Moments: The scatter plot shows that there is a point around 50 scoring where teams basically don't lose if they score that much. There is also a point around 50 scoring where they very rarely lose, with one away team this time.

VII. Hypotheses for the Study

Based on the visualizations and insights provided from the Exploratory Data Analysis (EDA) phase, several hypotheses can be formulated for the overall study. These hypotheses can either be verified through further statistical analysis or commented upon based on the trends observed in the data.

A. *Home-Field Advantage Hypothesis*

- Statement: Teams playing at home are more likely to win than teams playing away.
- Rationale: Figure 1-4's scatter plots and box plots concerning home scores versus game results show that when a team's home score increases, the probability that they'll win the game (result = 1) increases dramatically. The home team's win probability jumps to nearly 100% if they score more than 80 points. Meanwhile, the outcome pie chart in figure 6 reveals that 57.1% of the time, the home team wins, a statistic that obviously supports the idea of home-field advantage and which was one of the hypotheses being tested in the chapter.

B. *Offensive Performance Hypothesis*

- Statement: Teams that score more points are more likely to win, regardless of whether they are playing at home or away.
- Rationale: Both scatter plots (home and away scores vs. game result) fig.1 and fig.2 confirm that higher team scores by the home or away team are strongly correlated with winning outcomes. This shows that the importance of offensive performance as a key predictor of game results.

C. *Variability in Away Team Performance Hypothesis*

- Statement: Away teams exhibit greater variability in performance compared to home teams.
- Rationale: The scatter plot for the away scores is broader when compared to the home scores, which shows there is a greater variability for away team than the home team. Some of the away teams perform really good and win most of the games while the other struggle.

D. *Defensive Strength Hypothesis*

- Statement: Teams that allow fewer points (stronger defenses) are more likely to win.
- Rationale: The Features related to defensive performance, such as the points allowed by the home team in the third quarter(q3) over the last 10 games shows moderate correlations with game outcomes. This indicates that defensive strength is a main factor in determining success, particularly during crucial moments of the game, such as third downs or late-game situations.

E. *Critical Performance Moments Hypothesis*

- Statement: Teams that perform well in critical moments (e.g., 4th quarter or last 2 minutes) are more likely to win.
- Rationale: The features related to offensive performance, such as the home team's score in the fourth quarter (Q4) and the home team's score in the last two minutes of the fourth quarter over the last 10 games, show strong correlations with game outcomes. This suggests that late-game offensive performance is a main factor in determining success, particularly during high-pressure moments, when games are often decided.

F. *Momentum Hypothesis*

- Statement: Teams with recent wins tend to maintain momentum and perform better in subsequent games.
- Rationale: The feature related to the home performance, specifically the number of wins in the last 10 home games, shows a strong correlation with game outcomes. This suggests that teams with a history of recent victories tend to maintain their success for future performance.

VIII. Data Visualization and Results

We will cover the types of machine learning and deep learning models used in our project in this section. More specifically, we will concentrate on their architecture, hyperparameters, and results [10]. Additionally, the ELO rating system is also discussed as a traditional method for predicting NFL game outcomes. Each approach has its strengths and weaknesses, and more specifically, we will concentrate on their architecture, hyperparameters, and results.

A. *Random Forest Classifier (Machine Learning Model)*

Architecture and Hyperparameters

- We fine-tuned the Random Forest model by means of a grid search with cross-validation to find the hyperparameters that perform best. The best parameters we got with best results are:
  - n_estimators: 200
  - max_depth: 10
  - min_samples_split: 5
  - min_samples_leaf: 2
- These parameters improved the model's performance when we have tried with the test dataset.

Results:

- With hyperparameter tuning, the Random Forest model achieved a test accuracy of 64.2% using the top 25 features.
- The confusion matrix for the Random Forest model revealed that:
  - 110 home team wins are predicted correctly out of 204 home wins.
  - 211 wins are predicted correctly with total of 293 actual wins for away teams.
  - Ties remained challenging to predict as there are only less number of ties in actual dataset.

Key Observations:

- There is a high precision in predicting the home wins.
- For away team predictions there is moderate accuracy.
- As we have less number of ties in actual the performance is also low.

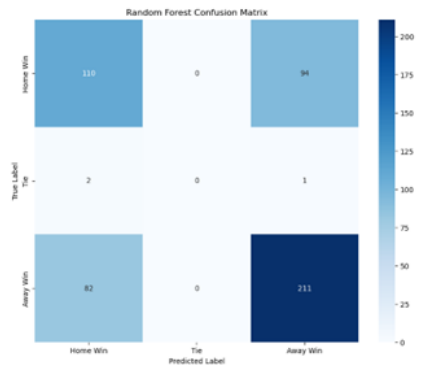

Fig. 8. Random Forest Confusion Matrix.

B. *Recurrent Neural Network (RNN) - Deep Learning Model*

Architecture and Hyperparameters

Recurrent Neural Network is a neural network specially made to deal with sequential type of data.Neural networks process input data and output will come as a transformation of their input, with the hidden layer in between that capture information about how they have performed previous time steps. In our project, we trained an RNN model with three layers to predict the outcome of a game based on sequential data from the last 10 games. The network architecture we have used in this model and optimization layers were similar to what was used for a football RNN by [6].

- Input Layer: The input layer consists of features derived from the last 10 games from our above analysis and aslso averages for both home and away teams.
- Hidden Layers: The RNN in our model consists of three hidden layers, each layer with 25 units. The activation function used here is tanh activation function, which helps to capture non-linear relationships between the features in our dataset and also while maintaining smooth gradients during backpropagation.

- Output Layer: The output layer consists of three units representing the three possible outcomes: home team win (0), tie (1), and away team win (2). A softmax activation function is used in the output layer to convert raw scores into output for each class(1,0.-1).
- The RNN model was tuned using a grid search method with the following hyperparameters:
  - hidden_size: [25, 50]
  - num_layers: [1, 2, 3]
  - dropout: [0.0, 0.2, 0.4]
  - learning_rate: [0.001, 0.01]
- The best configuration we got after tuning model is:
  - hidden_size: 25
  - num_layers: 2
  - dropout: 0.2
  - learning_rate: 0.001

Results:

- After hyperparameter tuning, the RNN model with above hyperparameters we have achieved a test accuracy of 64.2% , which equals the performance of the Random forest model.
- The confusion matrix for the RNN also shows similar trends as of the Random Forest model in terms of accuracy and precision with the slight variations:
  - There is a strong indicating performance in predicting the home wins.
  - While coming to the away wins performance 226 were predicted correctly out of 293 wins.
  - With limited data available of ties there is limited success.
- Key Observations:
  - From the below results we can predict the away teams wins more often and with little less deviation in home wins and less data about ties.

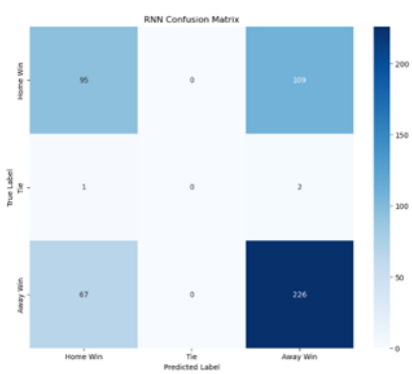

Fig. 9. RNN Confusion Matrix

Model Training and Performance Trends

- Below figure depicts the plots of the training loss and accuracy with the first 20 epoch accuracy increased but still the loss was more and gradually with 100 epochs both loss and accuracy almost becomes stable.
- Final training loss: 0.7411
- Final test loss: 0.6788

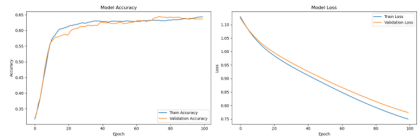

Fig. 10. Model Accuracy and Model loss.

RNNs can be very powerful for capturing temporal dependencies in sequential data. In this case, though, they didn't beat much simpler models like Random Forest. I could either be due to the dataset having insufficient sequential complexity or the hyperparameters like learning rate or batch size not being sufficiently tuned.

C. *Feature Selection: Top Features vs. All Features*

Feature Sets Evaluated

*1) **Top 10 Features:**:*

- We have selected top 10 based on correlation with game outcomes.
- Includes metrics like last_10_games_home_score, last_10_games_home_score_q4, and average_point_diff.
- With the top 10 features we have achieved an accuracy of 62% with Random Forest algorithm.

*2) **Top 25 Features:**:*

- Along with top 10 we have also Included additional metrics such as turnovers, time of possession, and defensive metrics (last_10_games_home_total_turnovers, last_10_games_away_yards_gained).
- Achieved an accuracy of 64.2%, outperforming both top 10 and all features.

*3) **All Features:**:*

- we have also trained the model by taking all numerical features from the dataset.
- Achieved an accuracy of 61%, indicating that including too many features there may be noise which reduces model performance.

Reason for Choosing Top 25 Features

when we used 25 features, it was mixture of all the relevant and essential features which caused proper fitting of the model. while using 10 features caused missing of key features which helped in predicting the outcomes. The key features which caused the difference were turnover differential and time of possession.

IX. ELO Rating System With Graph Database

A. *Overview*

The ELO rating system [8] is a traditional method for ranking players or teams based on their performances in direct matches against one another. In this project, ELO ratings were calculated for each team after every game, with ratings being updated based on actual game results versus expected results. We are using Neo4j- a graph database. In Neo4j, data

is stored as nodes and relationships. Each team that we are working with is represented as a node in the database, and the relationship between two nodes reflects the metrics of the game they just played, such as the final score and other relevant details. This leads to a unique Neo4j-based solution, which processes team ratings and enables their update after each game efficiently.

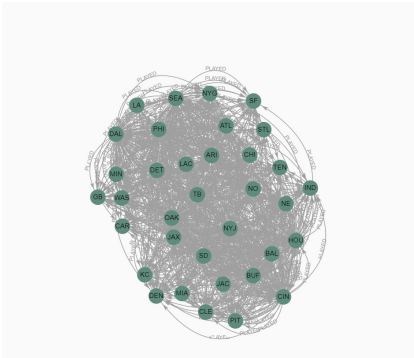

Fig. 11. Neo4j Database

B. *Calculation Process*

For each game:

- The official ELO ratings of both teams are obtained from the current standings.
- Expected win probabilities are calculated using the below formula:

$$E_A = \frac{1}{1 + 10^{\frac{R_B - R_A}{400}}}$$

where:

- EA is the expected probability of team A winning
- RA and RB are the current ratings of teams A and B respectively

The rating update formula used is:

$$R'_A = R_A + K(S_A - E_A)$$

where:

- K is the k-factor (set to 20 in our implementation)
- SA is the actual score (1 for win, 0.5 for tie, 0 for loss)
- EA is the expected score

The new ratings are saved in Neo4j graph data base each team corresponds to the node while each game is a relation between the nodes [9]

C. *Results*

The Elo rating system applied was able to guarantee a prediction accuracy of

- 63% on predicting outcomes for the last 500 games in the dataset.

This result shows that,it is possible to model the ELO ratings that reflects how stronger a team is becoming and makes rather reasonable predictions referring to the next games.

X. Conclusion and Future Scope

All three models: Random Forest, RNN, ELO, have their strengths:

- The accuracy of the Random Forest classifier stayed good with a smaller set of features and the use of simple feature engineering the model gives maximum possible accuracy around 64%.
- The RNN model did not perform better than simpler models, although it did capture sequential dependencies.
- The ELO rating system gave real-time flow of the team performance in the over time and the average accuracy of prediction was about 63%.

These results suggest us that combining traditional methods like ELO with modern machine learning techniques could help us to lead even better predictions in future iterations of this project.

In the Future work we can integrate the Elo ratings in to Machine learning pipelines by taking them as a additional input feature for the model.Moreover, investigating ensemble strategies that merge ELO with Random Forest could yield even better predictions, as these two methods complement each other well. This research avenue could lead to the development of the next generation of models that not only enhances but also pushes the limits of sports analytics.

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